

geo scoring algorithm

there are going to be different evaluators

1. citation authority evaluator
 2. evidence support evaluator
 3. readability evaluator
 4. entity coverage evaluator
 5. freshness evaluator
 6. FAQ/user intent evaluator
 7. structured data evaluator
 8. technical discoverability evaluator
-

based on this

1. citation authority evaluator

input

- Paragraphs
- External links

evaluates

- authoritative citations
- citation coverage
- credibility

score

0-5

2. evidence support evaluator

input

paragraphs

tables

find

- numbers
- percentages
- benchmarks
- measurements

evaluate: are they supported?

score

0-5

3. readability evaluator

input

headings

paragraphs

lists

evaluate

- clarity
 - logical flow
 - chunking
 - accessibility
-

4. entity coverage evaluator

input

paragraphs

title

headings

evaluate

- important entities
- terminology
- topic completeness

5. freshness evaluator

input

dates

content

schema

headers

evaluate

- updated recently
- outdated references
- stale information

6. faq / user intent evaluator

i actually dont think faq is the thing we want to evaluate

we are evaluating

| does this page answer the questions users or llms are likely to have?

a page can score perfectly without an actual faq section if it naturally covers those questions

7. Technical Discoverability Evaluator

Input

```
robots.txt  
json-ld  
sitemap  
meta  
llms.txt
```

this **doesnt need much semantic reasoning**

its mostly

- completeness
- correctness
- best practices

now, for each and every evaluator, we will have its separate rubric in its prompt so that it can understand what it has to and then do it

it is better to have the rubric as an operational rubric instead of having a rubric which is just gonna define a score

example, for citation evaluator

```
Step 1  
Identify all factual claims.  
  
Step 2  
Determine which require citations.  
  
Step 3  
Locate nearby citations.  
  
Step 4  
Evaluate source authority.
```

Step 5

Assign a score using the rubric.

scoring schema

"""

Score 5

- Nearly every factual claim has an authoritative citation.
- Sources are government, peer-reviewed, standards bodies, or official documentation.
- Citations directly support the adjacent claim.

Score 4

- Most factual claims cited.
- Small number of unsupported claims.

Score 3

- Some citations.
- Significant unsupported factual statements.

Score 2

- Few citations.

Score 1

- Rare citations.

Score 0

- No citations.

"""

Return JSON only.

algorithm

Evaluator score

$$E_i = (S_i / 5) * R_i$$

Trust & Authority score

$$\text{Trust} = ((E_{\text{citation}} + E_{\text{evidence}}) / 10) * 40$$

Content quality score

$$\text{Content} = ((E_{\text{readability}} + E_{\text{entity}} + E_{\text{freshness}} + E_{\text{faq}}) / 15) * 35$$

Technical discoverability score

$$\text{Technical} = ((E_{\text{schema}} + E_{\text{technical}}) / 7) * 25$$

Final GEO score

$$\text{GEO} = \text{Trust} + \text{Content} + \text{Technical}$$

Evaluator confidence

$$EC_i = C_i * R_i$$

Trust confidence

$$\text{TrustConf} = ((EC_{\text{citation}} + EC_{\text{evidence}}) / 10)$$

Content confidence

```
ContentConf = ((EC_readability + EC_entity + EC_freshness + EC_faq) / 15)
```

Technical confidence

```
TechnicalConf = ((EC_schema + EC_technical) / 7)
```

Overall evaluation confidence

```
OverallConfidence = 0.40 * TrustConf + 0.35 * ContentConf + 0.25 * TechnicalConf
```